$p := 10^{-12}$
$n := 10^{-9}$
$u := 10^{-6}$
$m := 10^{-3}$
$k := 10^{3}$
$M := 10^{6}$

$V_{IN} := 22.2$

$I_{OUT} := 1$

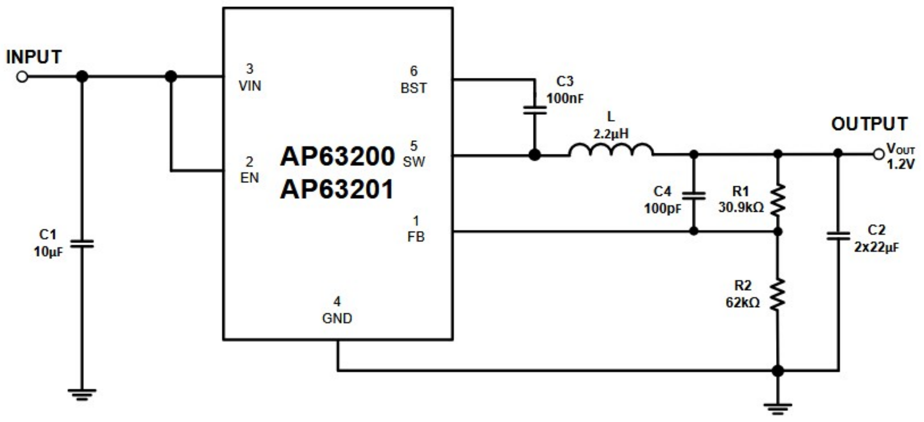

Figure 20. Typical Application Circuit of AP63200/AP63201

$$\mathbf{R_1} = R_2 \cdot \left(\frac{V_{OUT}}{0.8V} - 1\right)$$

$R_P := 50\ k$

$R_2 := 3.9\ k$

$R_{FIX} := 18\ k$

$$V_{OUT\_MIN} := 0.8 \cdot \left(\frac{R_{FIX}}{R_2} + 1\right) = 4.492$$

$$V_{OUT\_MAX} := 0.8 \cdot \left(\frac{R_{FIX} + R_P}{R_2} + 1\right) = 14.749$$

$$\mathbf{L} = \frac{V_{OUT} \cdot (V_{IN} - V_{OUT})}{V_{IN} \cdot \Delta I_L \cdot f_{sw}}$$

$f_{sw} := 500\ k$

$\Delta I_L := 0.4 \cdot 2$

$$L := \frac{V_{OUT\_MAX} \cdot (V_{IN} - V_{OUT\_MAX})}{V_{IN} \cdot \Delta I_L \cdot f_{sw}} = 12.376\ u$$

$$L := \frac{V_{OUT\_MIN} \cdot (V_{IN} - V_{OUT\_MIN})}{V_{IN} \cdot \Delta I_L \cdot f_{sw}} = 8.958\ u$$

$$I_{L_{PEAK}} = I_{LOAD} + \frac{\Delta I_L}{2}$$

$$I_{L\_PEAK} := I_{OUT} + \frac{\Delta I_L}{2} = 1.4$$